A-A-54002
10 October 1989

COMMERCIAL ITEM DESCRIPTION

BUR, BONE (Fast Cutting, Corrosion-Resisting Steel)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-B-37965 which is cancelled.

This commercial item description covers corrosion-resisting steel bone burs suitable for use with high torque rotary handpieces.

Salient characteristics:

Material. The burs shall be fabricated of Class 4, Type 440A, martensitic corrosion-resisting steel in accordance with ASTM Designation F899 for composition only. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

Hardness. After being properly hardened and passivated, the burs shall be stress relieved. The Rockwell hardness shall be C45 to C55 when tested in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Style, design and construction. The burs shall be of one piece construction, having a round head (operative end), and a "J" notched proximal end. The burs shall comform to Figure 1 for design and dimensions for the applicable National Stock Number.

Runout. The total indicated runout of the burs shall not exceed 0.003 inch when measured as specified herein.

Autoclavability. The burs shall not show evidence of corrosion when autoclaved at 270 ± 5 degrees (^{0}F) and 27 ± 1 pounds per square inch (psi) pressure for a minimum of 20 minutes as specified herein.

Corrosion resistance. The burs shall show no evidence of corrosion when subjected to the copper sulfate and boil tests as specified in ASTM Designation F1089.

AMSC N/A FSC 6515

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Finish. The head (operative end) shall have a honed finish with sharpened cutting edges for fast cutting. The neck and shank shall have a smooth finish. The shank shall be machine finished in accordance with Figure 1.

Identification marking. The type or placement of the marking shall not adversely affect or interfere with fit, function and operation of the bur with its associated handpiece. The use of a trademark is acceptable. The marking shall be distinguishable to the unaided eye. All markings shall be permanent and legible.

Workmanship. The burs shall have uniformly cut and formed blades and shall be free from defects which detract from their appearance or may impair their serviceability.

Unit. Each (EA). One bur, of the stock number specified, constitutes one unit.

Unit package. Each bur shall be packaged in a heat sealed bag and placed in an appropriate shipping container which prevents damage to the bur and has an adequate closure which prevents spillage of contents during shipping and handling.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level of II. Examination shall include, but shall not be limited to the defects listed below. The unit of product for examination purposes shall be one bur, as specified.

Defects

Bur size and shape not as specified.
Bur not of one piece construction.
Number of flutes not as specified for the applicable bur.
Cutoff end on shank not as shown in applicable figure.
Burs not free from defects such as chipped blades, cracks, or pits.
Finish on bur not as specified.
Instructions for use inaccurate, inadequate or missing.

Examination. The burs shall be examined to determined compliance with this description.

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Visual examination. Examination for posssible bur defects not perceptible to the unaided eye shall be made by using a 10X magnification.

Sampling for tests. Sampling for tests shall be conducted in acccrdance with MIL-STD-105 at inspection level S-2. The unit of product for test purposes shall be one bur. Where feasible, the same burs shall be used for two or more test characterisitics. A sufficient quantity of each lot of raw material used in the manufacture of the burs shall be selected for chemical analysis.

Tests. Tests shall be conducted to determine compliance with this commerical item description.

Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadowgraph or other suitable equipment of equal accuracy shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Autoclavability. The test bur shall be cleaned in accordance with the instructions supplied by the manufacturer. The bur shall then be autoclaved at 270 $\pm$ 5^{0}F and 27 $\pm$ 1 pounds per square inch pressure for a minimum of 20 minutes. After the bur is removed from the autoclave, the entire surface area of the bur shall be examined for evidence of corrosion.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6515-01-167-3788	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 3.2 mm Head Diameter, 6 Flutes, 44 mm Long
6515-01-167-3789	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 4.8 mm Head Diameter, 6 Flutes, 44 mm Long
6515-01-166-7786	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 4.8 mm Head Diameter, 6 Flutes, 55 mm Long
6515-01-166-7787	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 3.2 mm Head Diameter, 6 Flutes, 55 mm Long
6515-01-166-7788	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 6.3 mm Head Diameter, 8 Flutes, 44 mm Long
6515-01-166-9005	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 4.0 mm Head Diameter, 6 Flutes, 55 mm Long
6515-01-166-9006	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 2.4 mm Head Diameter, 4 Flutes, 44 mm Long
6515-01-167-2616	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 5.5 mm Head Diameter, 8 Flutes, 44 mm Long

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6515-01-167-2622	BUR, BONE, Fast Cutting, Corrosion-Resisting Steel, 4.0 mm Head Diameter, 6 Flutes, 44 mm Long

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6515-4716
LOCATION: NCR DISK 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-54002	BUR, BONE (Fast Cutting, (Corrosion-Resisting Steel

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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